

Agam Airi

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Software Engineer offering **3+ years** of expertise in **full-lifecycle development** and **low-level hardware integrations**. Proven track record in engineering **IoT ecosystems** and **enterprise architectures, open-source contributions**, bridging custom firmware with scalable mobile and web deployments. Focused on **clean code practices, unit/integration testing**, and shipping **production-ready software**.

EXPERIENCE

Mobile Application Developer Solaris Robots

February 2024 – Present
Oakville, ON

Mobile developer owning the full lifecycle of iOS and Android apps for robotics, IoT, and airport operations, from native plugin development to Playstore, App Store and Enterprise MDM release. Work cross-functionally with product managers in an Agile, Jira-driven process, with a focus on hardware integration, performance, and shipping reliable production software.

- Led the PRMGO platform migration to **Flutter and Node.js**, scaling it to **300+ concurrent Toronto Pearson airport staff** and **1,000+ daily passenger-tracking events**.
- Built a **proprietary Kotlin bridge** for **low-level UART communication**, enabling direct control of **custom Android hardware, battery modules, and ToF sensors**.
- Delivered **IoT device telemetry** and over-the-air **firmware updates** by engineering **Swift and Kotlin** bridges that **bypassed cross-platform limitations** for deep hardware integration.
- **Cut data and application load times by 30%** across SolarX UV-disinfection robotics apps by **profiling performance bottlenecks**, using **debugging tools** and resolving critical production bugs.
- Shipped releases across the **TestFlight, App Store, Google Play, and Enterprise MDM** by owning **full-cycle deployment** and managing **Agile workflows** in Jira.
- Sustained quality over **complex system architectures** by writing **unit and integration tests** and **technical documentation**, following **clean code practices, OOP fundamentals** and proper **state management**.
- Led **peer code review** of AI-generated code, **auditing security, structure, and integrity** to **flag vulnerabilities** and guide architectural fixes across the team.

Independent Software Developer & Mentor (Freelance)

May 2023 – January 2024
Toronto, ON (Remote)

Worked as a mobile developer on freelance and contract-style projects, while mentoring peers in core programming concepts. Developed open-source mobile and web applications, treating each build with production-grade architecture and delivery standards.

- **Shipped 10+ mobile and cross-platform** apps across **Flutter, Swift and React-** from a multi-platform surveillance app to native iOS utilities in Swift, SwiftUI and Objective C.
- Mentored **5+ junior developers** in **OOP principles** and **Data Structures**, facilitating measurable improvements in technical problem-solving and code quality.

PROJECTS

- **DreamAiri**: full-stack AI coding assistant for game development - [Website](#) - **GoLang (JWT, WebSocket)** backend with **PostgreSQL**, a **Next.js/TypeScript/Tailwind** portal deployed on **Firebase**, and a **GDScript editor plugin** - enabling multi-turn **agentic LLM workflows (OpenAI, Gemini, Ollama)** with per-user credit billing, license management, and real-time **tool execution** inside the Godot editor.
- **AI Council** : **Python** Multi-LLM Platform - [GitHub](#) - **Flask/SocketIO** platform where multiple local LLMs collaborate via sequential-debate, web-search, and deep-research modes, with document analysis and tool use - fully offline.
- **A.I.R.I** : **AI, Real-time, In-app** - [GitHub](#) - Privacy-first Flutter app running LLMs **locally (LLAMA.CPP)** with **on-device RAG**, multimodal chat (text/images), video and voice interaction (**TTS/STT**), and **LAN server mode** exposing an **OpenAI-compatible API**.
- **forge_mvvm** : Published Flutter Package · [pub.dev](#) · **200+ downloads** **MVVM + Clean Architecture framework** on pub.dev with **runtime dependency-graph validation, typed async primitives**, and a CLI that scaffolds complete feature modules.
- **PrepStation** : Flutter Video Editor · iOS + macOS - [GitHub](#) - Non-linear video editor in Flutter with a multi-track timeline, 30+ transitions, keyframe animation, FFmpeg export, and **real-time background removal powered by Apple Vision**.
- **MovieBrowser** : iOS App · Swift · SwiftUI - [GitHub](#) SwiftUI with **MVVM**, a **protocol-first** service layer, and Core Image poster-tinted gradients that adapt to light/dark mode. **Multi-layer caching (NSCache + URLCache)**, **Dependency Injection**, offline-aware **error states, Combine / async-await** and **XCTest** coverage across service, ViewModel, and persistence layers.

SKILLS

- **Languages**: Python, JavaScript/TypeScript, Kotlin, Swift, Dart, SQL
- **Web, Backend & Cloud**: Node.js, Express, REST APIs, Google Cloud (GCP), Firebase, Amazon SNS, MongoDB, SQLite, Docker
- **Frontend & System Integration**: React, HTML/CSS, Native OS Bridging (Platform Channels), Flutter, SwiftUI, Jetpack Compose
- **Architecture & Practices**: MVVM, Clean Architecture, Dependency Injection, State Management, Unit & Integration Testing, XCTest, CI/CD (GitHub Actions, Fastlane)
- **Data & Tools**: SQLite, MongoDB, SQL, Git, Xcode, Android Studio, Jira, Figma, Postman, Docker

EDUCATION

Associates Diploma of Computer Programming (2 years) - Seneca College, Toronto, ON
Bachelor of Business Administration with IT specialization - Sardar Patel University, India

2022-2023
2018-2021